05.06.2024

CSE2037 MICROPROCESSORS RE-SIT EXAM

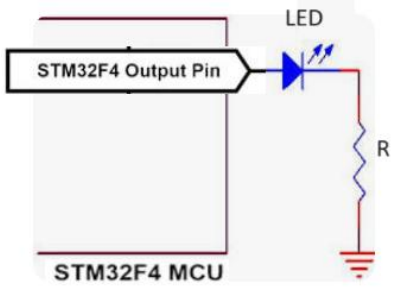
Name:

CLOSED BOOK, CLOSED NOTES EXAM. NO CALCULATORS OR CHEAT SHEETS ALLOWED.
PLEASE PROVIDE YOUR ANSWERS IN DESIGNATED PLACES.
DURATION: 60 min.

Q1) (15 points) Three microcontrollers (A, B & C) have maximum clock speeds 10MHz, 20MHz, 24MHz respectively. Microcontroller A divides its clock by 4 to give one machine cycle, microcontroller B by 8, and microcontroller C by 12. Microcontrollers A & C take 2 machine cycles to perform an instruction, while microcontroller B takes three cycles. Place the microcontrollers in order of the speed in which they can perform that instruction?

Q2) (15 points) You have a 12-bit, 0 to 3V range ADC. If the ADC input is 0.75 V, what will be the digital value in hex returned by this ADC?

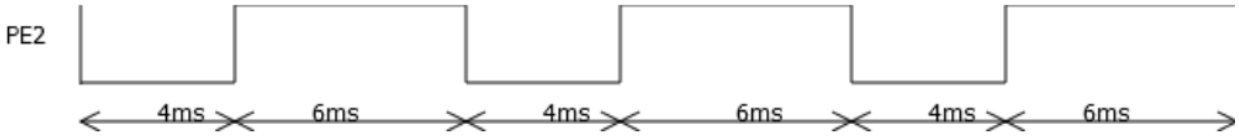
Q3) (15 points) Interface an LED to a selected output pin as given in the figure. The desired LED operating point is 2.4V, 1mA. The microcontroller output high voltage is 3.3V. Find required resistor value R.



Q4) (15 points) If R0 equals -10, what will be in register R0 after executing these instructions?

```
LSL   R1,R0,#3
ADD   R0,R0,R1
```

Q5) (30 points) When using the system timer, there is just one callback function named HAL_IncTick(void) for SysTick. This function is called every millisecond. Using SysTick Interrupt only to generate the following signal on Port E pin 2. (Write C code)



Q6) (20 points) A standalone embedded system is to be designed. As power source, a 1800 mAh battery is selected. Embedded system spends 10 seconds in run mode in every ten minutes and remaining time in sleep mode. Run and sleep modes draw 150 mA and 10 mA respectively. Calculate the time of battery change.